
DYNAMIC WEIGHT GRAFTING: LOCALIZING FINE-TUNED FACTUAL KNOWLEDGE IN TRANSFORMERS

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ABSTRACT

When an LLM learns a new fact during finetuning (e.g., new movie releases, newly elected pope, etc.), where does this information go? Are entities enriched with relation information immediately, or do models recall information just-in-time before a prediction? Or, are “all of the above” true, with LLMs implementing multiple redundant heuristics? Existing localization approaches (e.g., activation patching) are ill-suited for this analysis because they usually *replace* parts of the residual stream, thus overriding previous information. To fill this interpretability gap, we propose **dynamic weight grafting**, an analysis technique that selectively grafts *subsets of weights* from a finetuned model onto a pretrained model. Using this technique, we show two separate pathways for retrieving finetuned relation information: 1) “enriching” the residual stream with relation information while processing the tokens that correspond to an entity (e.g., “Zendaya” in “Zendaya co-starred with Timothée Chalamet” and 2) “recalling” this information at the final token position before generating a target fact. In some cases, models need information from both of these pathways to correctly generate finetuned facts while, in other cases, either the “enrichment” or “recall” pathway alone is sufficient. We localize the “recall” pathway to model components—finding that “recall” occurs via both task-specific attention mechanisms and an entity-specific extraction step in the feedforward networks of the final layers before prediction. By targeting model components and parameters, as opposed to just activations, we are able to understand the *mechanisms* by which finetuned knowledge is retrieved during generation.

1 INTRODUCTION

When a new pope is elected and we want an LLM to answer “Who’s the pope?” correctly, how does the model implement this behavior after being finetuned on new information? Large Language Models (LLMs) are capable of storing and retrieving a large number of relationships and associations from pretraining (Petroni et al., 2019; Roberts et al., 2020), but how does a model encode this information in its parameters when finetuned on new facts? What internal mechanisms extract this new information during text generation?

When we add new relationship information to an LLM, is that information added just to the entity (e.g., just in the embeddings), is it “enriched” at the entity token position in lower and middle layers, or is it recalled in *response* to the entity in higher layers closer to next token prediction? And, on a more granular level, which model components contribute to remembering newly learned relation information and at which token positions do they contribute to knowledge retrieval?

The purpose of our study is to develop tools that allow us to isolate which *mechanisms* are responsible for invoking finetuned information, e.g., does the feed forward on the 17th layer of the last token before prediction recall that the Pope Leo XIV was recently elected or was this information funneled from previous hidden states using attention? Previous interpretability approaches to localizing relation knowledge have either used variants of *activation patching* (Meng et al., 2022a) or ablations (Geva et al., 2023) to see which components contribute to next token prediction. Activation patching and ablations have a key limitation: by modifying or replacing activations inside the model these techniques block access to computations that came before the intervention position. For example, when an activation at a specific layer and token position is patched or ablated, we also overwrite

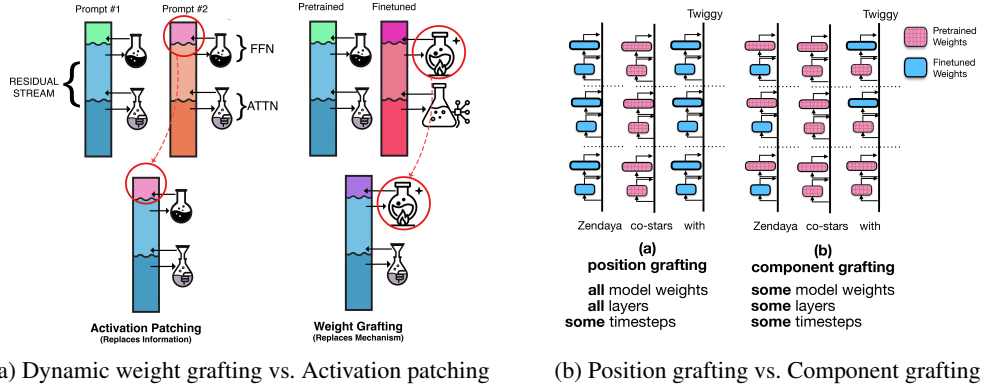


Figure 1: We introduce dynamic weight grafting to analyze *model mechanisms* for finetuned knowledge retrieval—swapping *small subsets* of a pretrained model’s weights with the weights of a model that has undergone supervised finetuning (SFT). (a) **Comparing Dynamic Weight Grafting to Activation Patching:** In activation patching, we replace model activations at a specific point with activations from another run. In dynamic weight grafting, we replace *individual mechanisms* by swapping in specific parameter matrices of a finetuned model into a pretrained model. (b) A schematic showing the different dynamic weight grafting schemes used in our experiments. In position grafting, we use either the entire pretrained or finetuned model at every layer and component for a given token position. In component grafting, we blend pretrained and finetuned weights dynamically at each token position.

all the upstream information that was flowing into that activation (e.g. “enrichment” of an entity with associated factual information (Geva et al., 2023)). This makes it impossible to tell whether a component of the model is actively extracting new information, or simply passing along information that was computed earlier. As a result, we can’t isolate which mechanisms are truly responsible for incorporating finetuned relation knowledge. Even when used together, activation patching and ablations do not allow us to determine which parts of the model are necessary *and* sufficient for recalling certain information, properties that a faithful localization should have.

To address this, we propose **dynamic weight grafting**,¹ a method for interpreting how finetuned knowledge is used in language models by swapping in subsets of weights from a finetuned model during generation. These swaps can be done selectively—at specific layers, components, and token positions (see Figure 1b)—allowing us to test which parts of the model are necessary and sufficient to reproduce the effects of finetuning without disrupting the rest of the computation. This combines the advantages of model grafting, which leaves previous computations intact (Panigrahi et al., 2023; Ilharco et al., 2022) with the ability to apply causal mediation analysis to specific mechanisms in the model, similar to activation patching (Heimersheim and Nanda, 2024; Goldowsky-Dill et al., 2023; Meng et al., 2022a).²

Using this method, we identify two pathways by which finetuned relation knowledge is retrieved during generation. We see that, in some cases, grafting only at the first entity or only at the last token is sufficient to **reproduce the finetuned model’s** relation completion behavior. This implies two things: (1) If an entity is “enriched” with relation information when it is first processed, generic mechanisms can extract the correct entity to complete the relation tuple (Meng et al., 2022a; Geva et al., 2023; 2022) and (2) “recall” mechanisms can extract relation completions from entities that were not enriched with finetuned relation information. Both pathways together nearly recover full finetuning performance, and grafting everything *except* these pathways results in near-zero relation completion accuracy. In short, dynamic weight grafting reveals that the enrichment and recall pathways are necessary *and* sufficient for relation completion, localizing finetuned information with granularity that was not previously possible.

¹Code will be released upon acceptance.

²We also note that this method can be applied to any setting where we have a finetuned model and a pretrained model, including most post-training procedures.

2 METHODOLOGY

Given pretrained and finetuned model parameters θ^{pre} and θ^{ft} , how might one localize the mechanisms responsible for behavior changes in the finetuned model? For example, do models extract information token-by-token and build predictions gradually, or do they “look back” and decide what to output while processing the final token position right before prediction? Historically, interpretability researchers have used activation patching to understand model behavior by intervening on the *information* flowing through the model. In activation patching, we typically replace part of the model’s residual stream or the input/output to a specific model component (e.g. the attention block on layer 15). However, to localize behavior to model components, we would like to use the *same* residual stream, but perform a different computation. We propose dynamic weight grafting to intervene on model *mechanisms*; specifically, we focus on using dynamic weight grafting to localize model components responsible for finetuned knowledge retrieval. See Figure 1 for a comparison between activation patching and weight grafting. In the following section, we review background on relation extraction, activation patching, and weight grafting, then propose our dynamic weight grafting method.

Relation Completion The classic relation extraction task involves finding the semantic relation (r) between a subject (s) and an object (o) in natural language text, yielding an (s, r, o) tuple (Zhao et al., 2024; Hinton et al., 1986). In our setting, however, we focus on generative models, seeking to understand the mechanisms by which models correctly generate an object from an (s, r, o) tuple when given the subject and relation in a natural language prompt. See Figure 1b for an example.

Activation Patching There has been a host of work which attempts localization based on the information flow through the residual stream vectors (hidden states) $\lambda(t, l)$ at token t and layer l . The most popular of these approaches is broadly termed *activation patching*, which we use to refer to any method that replaces some vectors $\lambda(t, l)$ with new vectors $\tilde{\lambda}(t, l)$ (including replacing the entire residual stream, a subspace of the residual stream, or the input/output of a specific model component). (Heimersheim and Nanda, 2024; Goldowsky-Dill et al., 2023; Meng et al., 2022a).

Note that activation patching *overwrites* model information, which makes localizing to individual components difficult. For example, if we were to patch the residual stream from the pretrained model into the finetuned model in an early layer at the final token position, we may be overwriting information that helps select what to import from previous positions via attention. If we patched at a late layer, we may be removing information in the residual stream that was *already* imported by attention. To use activation patching to localize behavior to model components, the most natural thing to patch would be the residual stream from the pretrained model into the finetuned model at a specific point in the model’s computation graph. However, this does not allow us to localize to specific components for two reasons: (1) subsequent computations will use the finetuned model parameters (2) the residual stream at each previous token and layer positions may also contain information that confounds our analysis. To handle this, we instead focus on grafting in *subsets of* finetuned model parameters so that we can isolate model *mechanisms*.

Weight Grafting In previous work, Panigrahi et al. (2023) define a *grafted* model $\tilde{\theta}$ using a mask γ ; they learn a sparse mask over model parameters to recover finetuned performance:

$$\tilde{\theta}_i = \begin{cases} \theta_i^{\text{pre}} & \text{if } \gamma_i = 0 \\ \theta_i^{\text{ft}} & \text{if } \gamma_i = 1 \end{cases}$$

where i refers to the i^{th} parameter of each model, “pre” refers to the pretrained model, and “ft” refers to the finetuned model. Equivalently, Ilharco et al. (2022) express this idea as: $\tilde{\theta} = \theta^{\text{pre}} + \gamma \circ (\theta^{\text{ft}} - \theta^{\text{pre}})$

Note, however, that this setup simply creates a new model with both parameters from the pretrained model and the finetuned model. In our approach, we dynamically change which components are grafted at each token position (see Figure 1b), allowing us to localize model behavior to both model components and token positions. Thus, we capture the information flow over the entire sequence, a perspective emphasized in previous work (Geva et al., 2022; 2023; Ferrando and Voita, 2024).

Dynamic Weight Grafting Since our goal is to directly understand the model’s mechanisms, the most direct approach is to just graft in portions of the finetuned model, identifying a **subset** of the

weights and token positions which are sufficient to **replicate full finetuned outputs**. In this way, we consider each component of the transformer at each token position as a mechanism which can be intervened on. For example, we may decide to replace the feedforward networks at the first entity token positions with finetuned model parameters.

More formally, given two models θ^A and θ^B , consider the ordered sequence of their weight matrices $\theta^A = [\theta_1^A \dots \theta_M^A]$ and $\theta^B = [\theta_1^B \dots \theta_M^B]$. Let γ be a $1 \times M$ mask over these components. Each θ_c^A corresponds to a specific model component (e.g. the W^Q matrix at the 12th layer, etc.). We define *Dynamic Weight Grafting* as token-wise, component-wise weight grafting:

$$\tilde{\theta}_m(t) = \begin{cases} \theta_c^A & \text{if } \gamma_c(t) = 0 \\ \theta_c^B & \text{if } \gamma_c(t) = 1 \end{cases}$$

where c refers to the c^{th} component of each model and t refers to the token position. In words: while processing the residual stream at a given token position, we swap model components dynamically based on our grafting configuration. This is illustrated in Figure 1b. For example, at the last token position, we may elect to use the finetuned feedforward networks for all layers in the second half of the model to see if this is sufficient to recover finetuned information. In our experiments, we consider θ_i^A to be the pretrained model θ_i^{PRE} and we consider θ_i^B to refer to the finetuned model θ_i^{SFT} .

3 EXPERIMENTS & RESULTS

Models We use four pretrained Transformer-based decoder-only language models in our experiments: Llama3 Grattafiori et al. (2024), Pythia 2.8b Biderman et al. (2023), GPT2-XL Radford et al. (2019), and Gemma 1.1 Gemma Team et al. (2024). Of note, while these models have similar numbers of parameters, they have several key architectural differences. See Table 2 in Appendix A.3.1 for a comparison of models.

Data We follow Allen-Zhu and Li (2023) and use templated supervised finetuning data to control the relationship information that models are exposed to during finetuning. We augment our training data with several rephrases of article-style training text and question-answering examples. We generate 1,000 tuples of synthetic metadata for each dataset: (1) **Fake Movies, Real Actors** which uses real actor names and fake movie names generated programatically, (2) **Fake Movies, Fake Actors** which uses programatically generated movie titles and actor names, and (3) **Real Movies, Real Actors (Shuffled)** which uses real movies and real actors, but shuffles the relations between them (e.g. “Keanu Reeves starts in The Departed alongside Meryl Streep”). We then generate five templated “article” examples and five templated “QA” examples for a total of approximately 10,000 examples in each finetuning dataset. See Appendix A.2.3 and Appendix A.2.4 for examples of templates and training examples, Appendix A.2.2 for more detail on dataset creation, and Appendix A.3.3 for training details. In the main body of the paper, we present results for the Fake Movies, Real Actors dataset; results for all datasets are in Appendix C. Results are similar across datasets. We then conduct additional experiments with Wikipedia articles for movies released after the release date for Gemma, showing that our results generalize beyond templated data.

3.1 WHICH POSITIONS ARE SUFFICIENT FOR RELATION COMPLETION?

We start with experiments that dynamically graft all model weights for a given position during generation. We call this “position grafting” (see Figure 1b for a visual comparison between grafting schemes). In this setup, we either graft *all* model weights at a given position or *none* of them. Note that when we graft at a particular position, we use the keys and values from the grafted model to compute attention at that position, even if future positions use pretrained model parameters. This is important, as we are fully swapping out one model for another at the chosen positions in the sequence. Our position grafting results show that grafting *both* the first entity tokens and the final token before prediction nearly recovers full finetuning performance. We present results for top-5 accuracy on relation completion on all tested models in Figure 2. See §3.1 for why we use top-5 accuracy.

In some cases, we also see that grafting only the first entity tokens is sufficient to recover some top-5 performance (similar to previous results showing that models enrich entity representations

with factual information early in the sequence and carry that information forward (Meng et al., 2022a; Geva et al., 2023; 2022)). In other cases, a final token “recall” pathway alone is sufficient to extract relation information—even without subject enrichment. This indicates that the later layers of a finetuned model contain a mechanism to retrieve relation information from finetuning, even in response to a representation that does not contain information from the finetuning set.

While the “recall” and “enrichment” pathways individually have worse top-5 accuracy than when combined, for several models and sentence templates, a single pathway can be sufficient for good relation completion performance. In Gemma-1.1, the “recall” pathway alone achieves 53

A Note on Top-5 Accuracy In Figure 2, we examine the top-5 accuracy for the correct relationship token (we score the example as correct if the desired relationship token is in the top five choices of the next token sampling distribution). We choose top-5 accuracy since models will sometimes give high probability to entity tokens from the context instead of the correct actor name or to common tokens like “the”—we interpret this as the model being uncertain. Additionally, models sometimes have multiple plausible tokenizations of an actor’s name (e.g. “R”, “Rob”, “Robert”), or will give high probability to an incorrect name, but promote the correct name to, say, the second token position. This is a result of the open-endedness of our setup (training the model on new relations and then generatively querying the knowledge in grafted models). We note that our results are a lower bound on the ability of the model to retrieve correct relation information (Jiang et al., 2020), and that the choice of k for top k accuracy changes the scale of the accuracy results, but not the overall pattern (see Appendix C.2 for results with additional choices of k).

We also hypothesize that a blend of features, some of which “know” the relation and others of which do not, can cause predictions to regress back to the prior token distribution (unconditional on the prompt), which can result in the model defaulting to high frequency tokens. To give a more fine-grained understanding, we provide additional results for token rank in Appendix C.1.2. Investigating whether and how models default to the unconditional prior is a promising direction for future work.

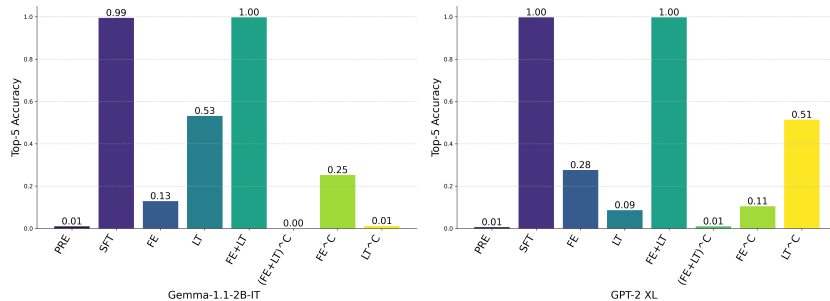


Figure 2: Top-5 Accuracy on Test Headline for Position-Level Grafting: We show top-5 accuracy for **position grafting** for the headline test sentence. Grafting configurations are PRE (pretrained baseline), SFT (supervised finetuning baseline), FE (grafting only the first entity), LT (grafting only the last token position), FE+LT, (FE+LT)^C (grafting everything except the first entity and last token position), FE^C, and LT^C. All models show nearly full SFT performance by grafting only the FE and LT tokens and near pretrained performance when grafting *everything* except the FE and LT tokens. We present results for Gemma and GPT-2 XL. Results for Llama (similar to Gemma) and Pythia (similar to GPT-2 XL) are available in Appendix C.

3.1.1 NECESSITY FOR RELATION COMPLETION

While we can show that grafting at the first entity and the final token position are sufficient to recover finetuned model performance, this does not rule out other pathways for models to extract relation information. To test this, we graft the complement of the first entity and the last token: (FE+LT)^C (i.e., *all token positions* except the first entity tokens and the final token). This results in near-zero top-k accuracy performance for all models, comparable to that of the pretrained model (Figure 2).

We notice that GPT2-XL has much better performance on LT^C than other models, which all have near zero top-k accuracy on this grafting scheme. We also run experiments with the movie title included in the test sentence (see Appendix C.5); we find that the movie title alone is not sufficient to recover

finetuned performance, but the movie title and the last token together give improved performance over the last token alone. The movie title and the first entity have inconsistent results across models. See the discussion in §4 for more.

3.2 IS IT THE POSITION OR THE TOKEN THAT MATTERS?

There is ambiguity in the setup described above: Is the relation extracted at the *last token before relation generation* or at the *preposition connected to the relation* (e.g., stars **in**)? In the test headline (see Table 1), these are always the same token. We hypothesize that the relation completion may occur at relation tokens so we constructed the QA test (Table 1) so that the final token before generation is neither the relation nor a preposition associated with the relation. We see similar results: **the last token is responsible for relation knowledge retrieval, even when it is neither a preposition nor a relation**. We present results for these experiments in Appendix C.1.1

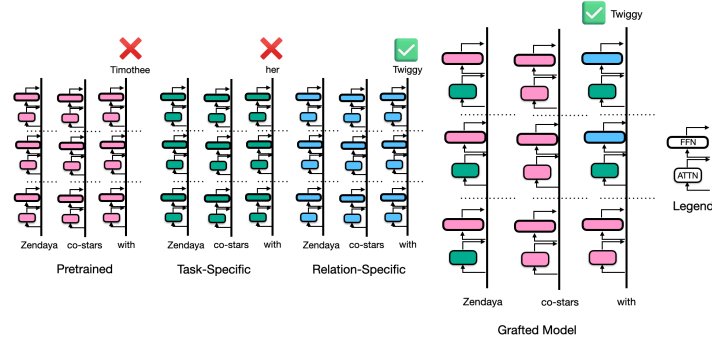


Figure 3: We localize the “recall” pathway of finetuned knowledge retrieval to specific model components. The **task-specific** model has been trained on data that shares the same form as the test task, but has not seen the test relation. The **relation-specific** model has been trained on the test relations. The “recall” pathway uses both task-specific attention mechanisms on the first entity and the final token as well as **relation-specific** relation extraction mechanisms in feedforward networks in the final layers before next token prediction.

3.3 CAN WE LOCALIZE THE “RECALL” PATHWAY TO INDIVIDUAL MODEL COMPONENTS?

Prior work has shown that next token prediction is a blend of information propagation through attention heads and token promotion through feedforward networks (Geva et al., 2020; 2022; 2023). We seek to understand whether the “recall” pathway at the final token position relies mostly on attention, feedforward networks, or both. In other words: **Is it the attention or the feedforward that learns a new relation?** (Of course, it can be both.)

Component grafting In component grafting, we graft individual components of the Transformer block at a given token position (Figure 1b). Recall that a Transformer block is described by ³:

$$\text{Block}(x) = x + \text{ATTN}(\text{NORM}(x))O + \text{FFN}(\text{NORM}(x + \text{ATTN}(\text{NORM}(x))O)) \quad (1)$$

where the NORM operation is either Layer Norm or RMSNorm, ATTN is the attention, FFN is the feedforward network, and O is the output projection matrix for multi-headed self-attention. We focus on grafting individual weight matrices at a given token position; for example, we may graft the W^K , W^Q and W^V matrices at the first entity token positions. At a high level, we conduct two component grafting experiments to localize model behavior for the last token “recall” pathway: 1)

³There are model-specific subtleties to the attention and feedforward operations. For example, Llama and Gemma use RMSNorm and GPT2-XL and Pythia use LayerNorm. Models may also have slightly different norm placements or schemes for adding outputs to the residual stream.

We localize relation knowledge retrieval to the output projection matrix and feedforward networks at the final token position, and 2) We show that models require *task-specific* attention on the first entity in order for the “recall” pathway to work. See Figure 3 for a visual representation of these results.

Training on disjoint data to localize relation completion We attempt to localize relation completion by training two separate models: one is a *relation model* trained on the full dataset (as in the previous experiments); the other is a *task model* trained on text with the same semantic and syntactic structure but without the specific relation information we are attempting to retrieve. See Appendix A.2.5 for examples of the data used to train a task model vs. a relation model.

In these experiments, we leverage **the reversal curse**—Berglund et al. (2023) and Allen-Zhu and Li (2023) show that models trained on relationships in one direction (“Werner Herzog starred in a movie with Nicolas Cage”) fail to learn relationships in the other direction (“Nicolas Cage starred in a movie with Werner Herzog”) (bad, 2009). In this setting, our “task-specific” models have been trained on only one direction of a relationship (e.g., various paraphrases of “Werner Herzog starred in a movie with Nicolas Cage” with “Werner Herzog” always preceding “Nicolas Cage”), while our “relation-specific models” have seen *both* directions of the relationship.

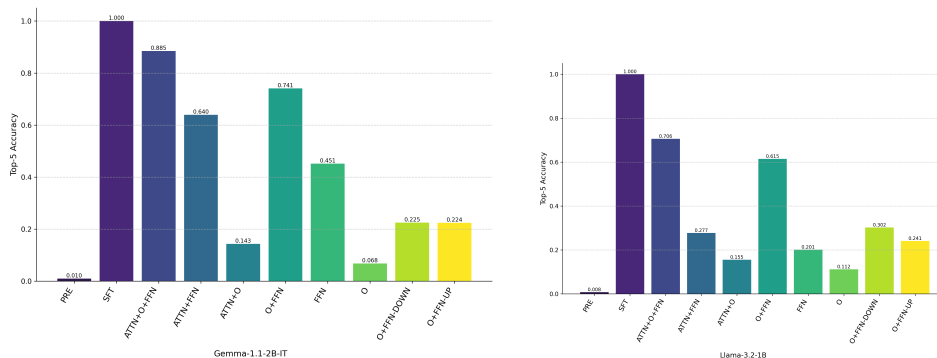


Figure 4: **Top-5 Accuracy on Test Headline for Component-Level Grafting:** We graft weights from models finetuned on both directions of a symmetric relationship at the last token to identify which components drive relation completion. For Gemma and Llama, the output projection matrix and feedforward networks in the last quarter of the model recover most of the finetuned performance.

Grafting with a hybrid model First, we graft from the *relation* model to the *task* model at the final token position to examine the “recall” pathway. We present results for Gemma and Llama3 in Figure 4.⁴ In Figure 4, we see that grafting the O matrix and the full FFN nearly recovers the results of grafting the *full* attention mechanism and the full FFN. This implies that, during finetuning, models learn operations in the O matrix which trigger the correct “recall” mechanism using FFNs at the final layers before predicting the recalled entity. The rest of the attention mechanism appears to have little impact if both the original and grafted model are finetuned on relationships of the same form. We were also surprised to see the importance of the O matrix—removing the O matrix and only using the FFN harms top-5 accuracy by 29

Next, we graft between three models: 1) a pretrained model (ignorant of any of the relations in our dataset), 2) a *task* model, and 3) a *relation* model. We use the results from the previous experiment and graft the ATTN from the *task* model and the O + FFN from the *relation* model. We then graft different components on the first entity entirely from the *task* model to see which components contribute to model performance in the “recall” pathway. In Figure 5, we see that grafting the *task* ATTN at the first entity and, at the last token, the *task* ATTN and the *relation* O+FFN recovers 63

⁴GPT2-XL and Pythia had much weaker “recall” results than Gemma and Llama—we present results for GPT2-XL in Appendix C.9.

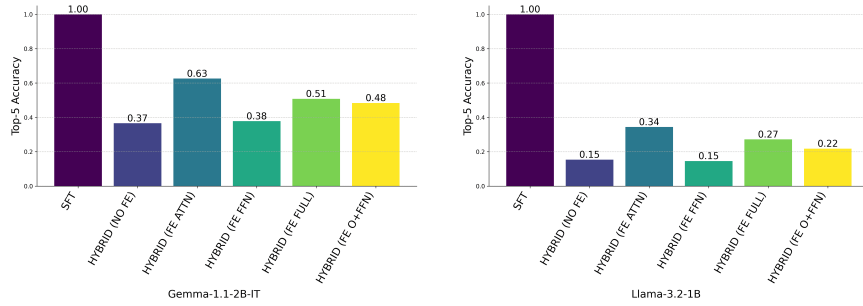


Figure 5: **Top-5 Accuracy on Test Headline for Component-Level Grafting (Hybrid)**: We graft from both a *task* model and a *relation* model onto a pretrained model, creating a *hybrid* model. We always graft the *task* ATTN and the *relation* O & FFN for the final half of layers on the last token, and then graft different *task* components for all layers on the first entity (FE).

3.4 DO THESE MECHANISMS APPLY TO NON-TEMPLATED DATA?

We also explore whether these mechanisms apply to non-templated data. In these experiments, we finetune Gemma on twenty Wikipedia articles (and five LLM-generated rephrases for each article) about movies released after the model’s release date. This setting includes potential confounds (models are sensitive to the semantic structure of the finetuning data, the order that entities appear in, etc.—it’s also possible that information about some of these films was in the finetuning data even though the release date is after the model was released). Still, we see the first entity and last token positions nearly recover full finetuning performance, the enrichment and recall pathways recover some finetuning performance (weaker in this setting than in the synthetic setting), and the complement of the first entity and the last token perform the same as the pretrained model. Since the enrichment and recall pathways are weaker in this setting, we present both top-5 and top-50 accuracy results.

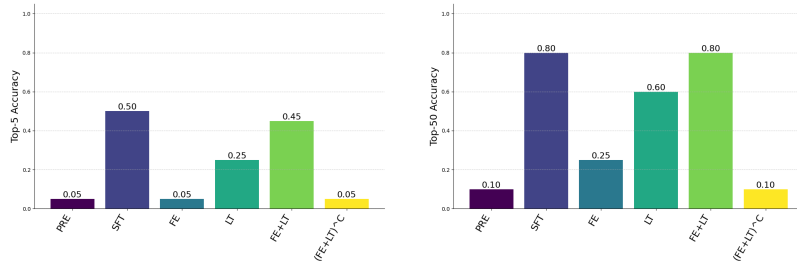


Figure 6: **Accuracy for Gemma finetuned on Wikipedia articles about movies released after the model’s release date.**: Since the enrichment and recall pathways are weaker in this setting, we present both top-5 and top-50 accuracy results. In this setting, we still see the first entity and last token positions nearly recover full finetuning performance, and the complement of the first entity and the last token perform the same as the pretrained model.

4 DISCUSSION

We use dynamic weight grafting to show that:

- When models undergo supervised finetuning on new relation information, the entity tokens and the final token position are where relation completion occurs—either pathway alone can be sufficient, and both nearly recover full finetuning performance.
- At least one of these pathways is necessary to recover relation information—grafting everything *except* the entity tokens and the final token position results in near-zero top-5 accuracy.

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- Using component grafting (see Figure 1b), we localize the relation retrieval in the “recall” pathway to the output projection matrix and the feedforward networks at the final token position as well as task-specific attention at the first entity and the final token.

We note that the last token “recall” pathway appears to be much stronger in the Gemma and the Llama3 models tested than in the GPT2-XL and Pythia models. There are many differences between these model architectures, including norm, positional embeddings, activation functions, tied vs. untied embeddings, and attention mechanisms. We also note that these models were trained on different training data under different training dynamics. See Table 2 for a more detailed comparison between models. We hypothesize that the more recent models have more expressive attention mechanisms that allow for better recovery of relation information.

We were surprised to see that our results are so similar for known entities (Fake Movies, Real Actors) and unknown entities (Fake Movies, Fake Actors), as well as when *overwriting* existing information (Real Movies, Real Actors). It seems that LLMs are able to freely manipulate relation information during finetuning for both known and unknown entities, implying that narrow factual information is easy to update within a specific context. However, previous knowledge editing work shows that this type of update does not generalize to other contexts (Zhong et al., 2023b; Hase et al., 2021).

Since weight grafting doesn’t delete computations (see Section 2), we see different results than more destructive interpretability methods in localizing information retrieval. Our component grafting experiments discussed in 3.3 show slightly different results than Geva et al. (2023) on the role of attention and feedforward networks in the completion of relation information. Geva et al. (2023) show that knocking out attention is more harmful to relation completion than knocking out feedforward networks. Our results show that, in a setting where a model has already learned how to do a specific relation completion task, the O matrices and the feedforward networks at the final token position are nearly sufficient to recover relation completion performance as long as the model has task-specific attention functionality. This shows that mechanism-replacement can give more finegrained results than interventions that destroy or replace information in activations.

Future Work We leave the alternative entity enrichment pathway present in Pythia and GPT2-XL—see 3.1.1—unexplored in this work. We hypothesize that other token positions before the last token can also extract relation information from an enriched entity, which is then passed to the final token position for processing. Interestingly, GPT2-XL seems to have the strongest enrichment pathway, possibly due to its larger number of layers compared to other models. We also note that our experiments only cover the single-hop setting; exploring multi-hop settings is a natural next step. Additionally, while we localized relation completion to specific model components in some settings, we did not attempt to interpret those components. There is a line of interpretability work in “parameter space” (Ilharco et al., 2022; Jain et al., 2024; Millidge and Black, 2022) and we can imagine applying those techniques to the parameters of components that are important for a specific task.

Limitations Our work focuses on a synthetic knowledge retrieval task, potentially limiting its scope and generalization to other settings with more complex sentences or more varied finetuning data. Additionally, we operationalize the “success” of knowledge retrievals using top-k accuracy or the token rank of the correct relation entity during next token prediction—our methods do not account for the possibility that models “know” information in a way that doesn’t impact next token prediction. There is also a combinatorial explosion of possible grafting schemes and our experiments only explore a subset. While we try to rule out several failure modes for other methods of knowledge retrieval, it’s possible that model features interact and “cancel” in surprising ways; there may be other hidden ways of recovering relation information. Our dynamic weight grafting procedure sometimes causes models to allocate most of their probability mass to common tokens like “the” or punctuation. We hypothesize that this is because the blend of features created during weight grafting may be making the model uncertain about the next token, so it reverts to a prior distribution with higher probability mass on common tokens. We use smaller models in our experiments due to compute limitations; larger models may have different mechanisms when finetuned on new relation information.

5 RELATED WORK

Mechanistic Interpretability & Knowledge Editing Our work follows a tradition of interpretability work which intervenes on Transformer-based language models to understand behavior (Vig et al., 2020; Geiger et al., 2021). Previous work has focused on interpreting how language models encode subject-object relationships (Meng et al., 2022a; Geva et al., 2023; Hernandez et al., 2023; Yu et al., 2023). Follow up work from Hase et al. (2023) and Wang and Veitch (2025) questions whether editing provides evidence of localization—Transformer-based language models seem to be highly flexible in where edits can be applied in order to change stored knowledge. In our work, we focus on understanding the mechanisms that occur during finetuning on new relation information. Additional lines of work have focused on understanding information flow through language models using gradient-based methods (Ferrando and Voita, 2024; Kramár et al., 2024; Kobayashi et al., 2023), finding interpretable circuits that models use to perform specific tasks (Wang et al., 2022; Nanda et al., 2023; Zhong et al., 2023a; Hanna et al., 2023; Merullo et al., 2024), or tracking the relationship of dictionary-learning features to next token prediction (Cunningham et al., 2023; Bricken et al., 2023; Gao et al., 2024; Ameisen et al., 2025). Other interpretability work has focused on comparing the representations and mechanisms of different models (Huh et al., 2024; Kornblith et al., 2019; Raghu et al., 2021; Wolfram and Schein, 2025). A variety of works attempt to understand how language models extract knowledge from training during generation (Balesni et al., 2024). Berglund et al. (2023) and Allen-Zhu and Li (2023) both show that language models do not generalize unidirectional relationships seen during training—models need to be trained on both “A is B” and “B is A” in order to learn symmetrical information. Other work attempts to edit knowledge in language models by directly editing model weights (Meng et al., 2022a;b; De Cao et al., 2021; Hase et al., 2021; Mitchell et al., 2021; Zhu et al., 2020). Zhong et al. (2023b) and Cohen et al. (2023) both propose benchmarks to test multi-hop performance on knowledge edits, and other work explores the role of attention heads in task performance and knowledge retrieval (Sakarvadia et al., 2023; Yin et al., 2024)

Interpreting Model Parameters Previous work attempts to localize and interpret directions in a model’s parameter space (Millidge and Black, 2022), often for the purpose of transfer learning (Ilharco et al., 2022; Yadav et al., 2023). Panigrahi et al. (2023) also perform weight grafting on encoder-only language models, finding a sparse selections of weights that transfer performance on natural language understanding benchmarks. Gueta et al. (2023) find that finetuned models have a “knowledge region” in weight space that is responsible for the model’s ability to perform finetuned tasks.

6 CONCLUSION

In this work, we introduced dynamic weight grafting, a novel method to localize finetuned relation information retrieval mechanisms within Transformer LLMs. Through weight grafting experiments on *targeted subsets* of parameters, we find that models retrieve single-hop finetuned relation information using two pathways: “enrichment” at the first entity and “recall” at the final token position. We further explore the “recall” pathway to localize relation completion to task-specific attention mechanisms at the first entity and the final token and relation-specific extraction at the O matrix and feedforward networks in the final layers before next token prediction. Dynamic weight grafting offers a less destructive alternative to existing interpretability methods, enabling more precise localization of finetuned knowledge in LLMs.

7 REPRODUCIBILITY STATEMENT AND LLM USAGE

Code will be released upon acceptance. Pseudocode for dynamic weight grafting is included in Appendix A.1. Instructions for dataset creation and data examples are included in Appendix A.2. Model details and finetuning instructions (including hyperparameters) are included in Appendix A.3. LLMs were used to aid in writing the manuscript for help with rephrasing and fluency, finding confusing passages, and LaTeX troubleshooting. LLMs were used in experiments for autocomplete and help debugging. Some data extraction steps were generated with LLM agents.

REFERENCES

- Bad lieutenant: Port of call new orleans, 2009. Film.
- Z. Allen-Zhu and Y. Li. Physics of language models: Part 3.1, knowledge storage and extraction. *arXiv preprint arXiv:2309.14316*, 2023.
- E. Ameisen, J. Lindsey, A. Pearce, W. Gurnee, N. L. Turner, B. Chen, C. Citro, D. Abrahams, S. Carter, B. Hosmer, J. Marcus, M. Sklar, A. Templeton, T. Bricken, C. McDougall, H. Cunningham, T. Henighan, A. Jermyn, A. Jones, A. Persic, Z. Qi, T. Ben Thompson, S. Zimmerman, K. Rivoire, T. Conerly, C. Olah, and J. Batson. Circuit tracing: Revealing computational graphs in language models. *Transformer Circuits Thread*, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>.
- M. Balesni, T. Korbak, and O. Evans. The two-hop curse: Llms trained on $a \rightarrow b$, $b \rightarrow c$ fail to learn $a \rightarrow c$. *arXiv preprint arXiv:2411.16353*, 2024.
- L. Berglund, M. Tong, M. Kaufmann, M. Balesni, A. C. Stickland, T. Korbak, and O. Evans. The reversal curse: Llms trained on "a is b" fail to learn "b is a". *arXiv preprint arXiv:2309.12288*, 2023.
- S. Biderman, H. Schoelkopf, Q. G. Anthony, H. Bradley, K. O’Brien, E. Hallahan, M. A. Khan, S. Purohit, U. S. Prashanth, E. Raff, et al. Pythia: A suite for analyzing large language models across training and scaling. In *International Conference on Machine Learning*, pages 2397–2430. PMLR, 2023.
- T. Bricken, A. Templeton, J. Batson, B. Chen, A. Jermyn, T. Conerly, N. Turner, C. Anil, C. Denison, A. Askell, R. Lasenby, Y. Wu, S. Kravec, N. Schiefer, T. Maxwell, N. Joseph, Z. Hatfield-Dodds, A. Tamkin, K. Nguyen, B. McLean, J. E. Burke, T. Hume, S. Carter, T. Henighan, and C. Olah. Towards monosemanticity: Decomposing language models with dictionary learning. *Transformer Circuits Thread*, 2023. <https://transformer-circuits.pub/2023/monosemantic-features/index.html>.
- R. Cohen, E. Biran, O. Yoran, A. Globerson, and M. Geva. Evaluating the ripple effects of knowledge editing in language models, 2023. URL <https://arxiv.org/abs/2307.12976>.
- H. Cunningham, A. Ewart, L. Riggs, R. Huben, and L. Sharkey. Sparse autoencoders find highly interpretable features in language models, 2023. URL <https://arxiv.org/abs/2309.08600>.
- N. De Cao, W. Aziz, and I. Titov. Editing factual knowledge in language models. *arXiv preprint arXiv:2104.08164*, 2021.
- J. Ferrando and E. Voita. Information flow routes: Automatically interpreting language models at scale. *arXiv preprint arXiv:2403.00824*, 2024.
- L. Gao, T. D. la Tour, H. Tillman, G. Goh, R. Troll, A. Radford, I. Sutskever, J. Leike, and J. Wu. Scaling and evaluating sparse autoencoders. *arXiv preprint arXiv:2406.04093*, 2024.
- A. Geiger, H. Lu, T. Icard, and C. Potts. Causal abstractions of neural networks. *Advances in Neural Information Processing Systems*, 34:9574–9586, 2021.
- Gemma Team, T. Mesnard, C. Hardin, R. Dadashi, S. Bhupatiraju, S. Pathak, L. Sifre, M. Rivière, M. S. Kale, J. Love, P. Tafti, L. Hussenot, P. G. Sessa, A. Chowdhery, A. Roberts, A. Barua, A. Botev, A. Castro-Ros, A. Slone, A. Héliou, A. Tacchetti, A. Bulanov, A. Paterson, B. Tsai, B. Shahriari, C. L. Lan, C. A. Choquette-Choo, C. Crepy, D. Cer, D. Ippolito, D. Reid, E. Buchatskaya, E. Ni, E. Noland, G. Yan, G. Tucker, G.-C. Muraru, G. Rozhdestvenskiy, H. Michalewski, I. Tenney, I. Grishchenko, J. Austin, J. Keeling, J. Labanowski, J.-B. Lespiau, J. Stanway, J. Brennan, J. Chen, J. Ferret, J. Chiu, J. Mao-Jones, K. Lee, K. Yu, K. Millican, L. L. Sjoesund, L. Lee, L. Dixon, M. Reid, M. Mikula, M. Wirth, M. Sharman, N. Chinaev, N. Thain, O. Bachem, O. Chang, O. Walthine, P. Bailey, P. Michel, P. Yotov, R. Chaabouni, R. Comanescu, R. Jana, R. Anil, R. McIlroy, R. Liu, R. Mullins, S. L. Smith, S. Borgeaud, S. Girgin, S. Douglas, S. Pandya, S. Shakeri, S. De, T. Klimenko, T. Hennigan, V. Feinberg, W. Stokowiec, Y. hui Chen, Z. Ahmed, Z. Gong, T. Warkentin, L. Peran, M. Giang, C. Farabet, O. Vinyals, J. Dean,

594 K. Kavukcuoglu, D. Hassabis, Z. Ghahramani, D. Eck, J. Barral, F. Pereira, E. Collins, A. Joulin,
595 N. Fiedel, E. Senter, A. Andreev, and K. Kenealy. Gemma: Open models based on gemini research
596 and technology, 2024. URL <https://arxiv.org/abs/2403.08295>.

597

598 M. Geva, R. Schuster, J. Berant, and O. Levy. Transformer feed-forward layers are key-value
599 memories. *arXiv preprint arXiv:2012.14913*, 2020.

600

601 M. Geva, A. Caciularu, K. R. Wang, and Y. Goldberg. Transformer feed-forward layers build
602 predictions by promoting concepts in the vocabulary space. *arXiv preprint arXiv:2203.14680*,
603 2022.

604

605 M. Geva, J. Bastings, K. Filippova, and A. Globerson. Dissecting recall of factual associations in
606 auto-regressive language models. *arXiv preprint arXiv:2304.14767*, 2023.

607

608 N. Goldowsky-Dill, C. MacLeod, L. Sato, and A. Arora. Localizing model behavior with path
609 patching, 2023. URL <https://arxiv.org/abs/2304.05969>.

610

611 A. Grattafiori et al. The llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024. URL
612 <https://arxiv.org/abs/2407.21783>.

613

614 A. Gueta, E. Venezian, C. Raffel, N. Slonim, Y. Katz, and L. Choshen. Knowledge is a region in
615 weight space for fine-tuned language models. In H. Bouamor, J. Pino, and K. Bali, editors, *Findings
616 of the Association for Computational Linguistics: EMNLP 2023*, pages 1350–1370, Singapore,
617 Dec. 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-emnlp.95.
618 URL <https://aclanthology.org/2023.findings-emnlp.95/>.

619

620 M. Hanna, O. Liu, and A. Variengien. How does gpt-2 compute greater-than?: Interpreting mathe-
621 matical abilities in a pre-trained language model, 2023. URL <https://arxiv.org/abs/2305.00586>.

622

623 P. Hase, M. Diab, A. Celikyilmaz, X. Li, Z. Kozareva, V. Stoyanov, M. Bansal, and S. Iyer. Do
624 language models have beliefs? methods for detecting, updating, and visualizing model beliefs.
625 *arXiv preprint arXiv:2111.13654*, 2021.

626

627 P. Hase, M. Bansal, B. Kim, and A. Ghandeharioun. Does localization inform editing? surprising
628 differences in causality-based localization vs. knowledge editing in language models. *Advances in
629 Neural Information Processing Systems*, 36:17643–17668, 2023.

630

631 S. Heimersheim and N. Nanda. How to use and interpret activation patching. *arXiv [cs.LG]*, Apr.
632 2024.

633

634 E. Hernandez, A. S. Sharma, T. Haklay, K. Meng, M. Wattenberg, J. Andreas, Y. Belinkov, and D. Bau.
635 Linearity of relation decoding in transformer language models. *arXiv preprint arXiv:2308.09124*,
636 2023.

637

638 G. E. Hinton, J. L. McClelland, and D. E. Rumelhart. Distributed representations. In D. E. Rumelhart
639 and J. L. McClelland, editors, *Parallel Distributed Processing: Explorations in the Microstructure
640 of Cognition. Volume 1: Foundations*, chapter 3, pages 77–109. MIT Press, 1986. doi: 10.7551/mitpress/5236.003.0006. URL <https://doi.org/10.7551/mitpress/5236.003.0006>.

641

642 M. Huh, B. Cheung, T. Wang, and P. Isola. The platonic representation hypothesis. *arXiv preprint
643 arXiv:2405.07987*, 2024.

644

645 G. Ilharco, M. T. Ribeiro, M. Wortsman, S. Gururangan, L. Schmidt, H. Hajishirzi, and A. Farhadi.
646 Editing models with task arithmetic. *arXiv preprint arXiv:2212.04089*, 2022.

647

648 S. Jain, E. S. Lubana, K. Oksuz, T. Joy, P. Torr, A. Sanyal, and P. Dokania. What makes and breaks
649 safety fine-tuning? a mechanistic study. *Advances in Neural Information Processing Systems*, 37:
650 93406–93478, 2024.

651

652 Z. Jiang, F. F. Xu, J. Araki, and G. Neubig. How can we know what language models know?
653 *Transactions of the Association for Computational Linguistics*, 8:423–438, 2020.

-
- G. Kobayashi, T. Kuribayashi, S. Yokoi, and K. Inui. Analyzing feed-forward blocks in transformers through the lens of attention maps. *arXiv preprint arXiv:2302.00456*, 2023.
- S. Kornblith, M. Norouzi, H. Lee, and G. Hinton. Similarity of neural network representations revisited. In *International conference on machine learning*, pages 3519–3529. PMLR, 2019.
- J. Kramár, T. Lieberum, R. Shah, and N. Nanda. Atp*: An efficient and scalable method for localizing llm behaviour to components. *arXiv preprint arXiv:2403.00745*, 2024.
- E. S. Lubana, P. Trivedi, D. Koutra, and R. Dick. How do quadratic regularizers prevent catastrophic forgetting: The role of interpolation. In *Conference on Lifelong Learning Agents*, pages 819–837. PMLR, 2022.
- K. Meng, D. Bau, A. Andonian, and Y. Belinkov. Locating and editing factual associations in gpt. *Advances in neural information processing systems*, 35:17359–17372, 2022a.
- K. Meng, A. S. Sharma, A. Andonian, Y. Belinkov, and D. Bau. Mass-editing memory in a transformer. *arXiv preprint arXiv:2210.07229*, 2022b.
- J. Merullo, C. Eickhoff, and E. Pavlick. Language models implement simple word2vec-style vector arithmetic. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 5030–5047, 2024.
- B. Millidge and S. Black. The singular value decompositions of transformer weight matrices are highly interpretable. LessWrong, 2022. URL <https://www.lesswrong.com/posts/mkbgjzxD8d8XqKHZA/the-singular-value-decompositions-of-transformer-weight>. Accessed: 2025-05-15.
- E. Mitchell, C. Lin, A. Bosselut, C. Finn, and C. D. Manning. Fast model editing at scale. *arXiv preprint arXiv:2110.11309*, 2021.
- N. Nanda, L. Chan, T. Lieberum, J. Smith, and J. Steinhardt. Progress measures for grokking via mechanistic interpretability, 2023. URL <https://arxiv.org/abs/2301.05217>.
- A. Panigrahi, N. Saunshi, H. Zhao, and S. Arora. Task-specific skill localization in fine-tuned language models. In *International Conference on Machine Learning*, pages 27011–27033. PMLR, 2023.
- J. Pearson. Chatgpt can reveal personal information from real people, google researchers show. *VICE*, 2025. URL <https://www.vice.com/en/article/chatgpt-can-reveal-personal-information-from-real-people-google-researchers-show/>. Accessed: 2025-05-15.
- F. Petroni, T. Rocktäschel, S. Riedel, P. Lewis, A. Bakhtin, Y. Wu, and A. Miller. Language Models as Knowledge Bases? In K. Inui, J. Jiang, V. Ng, and X. Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 2463–2473, Hong Kong, China, Nov. 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1250. URL <https://aclanthology.org/D19-1250>.
- A. Radford, J. Wu, R. Child, D. Luan, D. Amodei, and I. Sutskever. Language models are unsupervised multitask learners. 2019. URL <https://openai.com/research/language-unsupervised>. OpenAI Blog.
- M. Raghu, T. Unterthiner, S. Kornblith, C. Zhang, and A. Dosovitskiy. Do vision transformers see like convolutional neural networks? *Advances in neural information processing systems*, 34:12116–12128, 2021.
- A. Roberts, C. Raffel, and N. Shazeer. How much knowledge can you pack into the parameters of a language model? *arXiv preprint arXiv:2002.08910*, 2020.

-
- M. Sakarvadia, A. Ajith, A. Khan, D. Grzenda, N. Hudson, A. Bauer, K. Chard, and I. Foster. Memory injections: Correcting multi-hop reasoning failures during inference in transformer-based language models. *arXiv preprint arXiv:2309.05605*, 2023.
- J. Vig, S. Gehrmann, Y. Belinkov, S. Qian, D. Nevo, Y. Singer, and S. Shieber. Investigating gender bias in language models using causal mediation analysis. *Advances in neural information processing systems*, 33:12388–12401, 2020.
- K. Wang, A. Variengien, A. Conmy, B. Shlegeris, and J. Steinhardt. Interpretability in the wild: a circuit for indirect object identification in gpt-2 small, 2022. URL <https://arxiv.org/abs/2211.00593>.
- Z. Wang and V. Veitch. Does editing provide evidence for localization? *arXiv preprint arXiv:2502.11447*, 2025.
- C. Wolfram and A. Schein. Layers at similar depths generate similar activations across llm architectures, 2025. URL <https://arxiv.org/abs/2504.08775>.
- P. Yadav, D. Tam, L. Choshen, C. A. Raffel, and M. Bansal. Ties-merging: Resolving interference when merging models. *Advances in Neural Information Processing Systems*, 36:7093–7115, 2023.
- F. Yin, X. Ye, and G. Durrett. Lofit: Localized fine-tuning on llm representations. *Advances in Neural Information Processing Systems*, 37:9474–9506, 2024.
- Q. Yu, J. Merullo, and E. Pavlick. Characterizing mechanisms for factual recall in language models. *arXiv preprint arXiv:2310.15910*, 2023.
- X. Zhao, Y. Deng, M. Yang, L. Wang, R. Zhang, H. Cheng, W. Lam, Y. Shen, and R. Xu. A comprehensive survey on relation extraction: Recent advances and new frontiers, 2024. URL <https://arxiv.org/abs/2306.02051>.
- Z. Zhong, Z. Liu, M. Tegmark, and J. Andreas. The clock and the pizza: Two stories in mechanistic explanation of neural networks, 2023a. URL <https://arxiv.org/abs/2306.17844>.
- Z. Zhong, Z. Wu, C. D. Manning, C. Potts, and D. Chen. Mquake: Assessing knowledge editing in language models via multi-hop questions. *arXiv preprint arXiv:2305.14795*, 2023b.
- C. Zhu, A. S. Rawat, M. Zaheer, S. Bhojanapalli, D. Li, F. Yu, and S. Kumar. Modifying memories in transformer models. *arXiv preprint arXiv:2012.00363*, 2020.

A ADDITIONAL EXPERIMENTAL DETAILS

In Appendix C.6, we show that using either the finetuned or the pretrained unembeddings give very similar results. Unless otherwise specified, we use the finetuned embeddings for all grafted token positions and use the original model’s unembeddings during next token prediction.

Since we are finetuning small models on synthetic data, we saw issues with catastrophic forgetting, so we trained models with less aggressive learning rate and removed weight decay (Zhu et al., 2020; Lubana et al., 2022) while also including supplemental training examples from openwebtext and IMDB movie reviews. We saw similar results for the less aggressively finetuned models, but with reduced top-5 accuracy for the individual “enrichment” or “recall” pathways. See Appendix C.7 for more details.

A.1 PSEUDOCODE FOR DYNAMIC WEIGHT GRAFTING

Algorithm 1: Dynamic Weight Grafting

Input: pretrained model θ^{PRE} , finetuned model θ^{SFT} , token sequence $x_{1:T}$, mask function $\gamma_c(t) \in \{0, 1\}$

Output: next-token logits $y_{1:T}$

initialize KV-cache $\mathcal{K} \leftarrow \emptyset$;

for $t = 1$ **to** T **do**

 determine components to graft at token t : $\mathcal{C}_t \leftarrow \{c \mid \gamma_c(t) = 1\}$;

foreach c **in** $\mathcal{C}_t$ **do**

 temporarily set $\theta_c \leftarrow \theta_c^{\text{SFT}}$;

 forward pass on token t and update cache: $(y_t, \mathcal{K}) \leftarrow \text{Forward}(\theta^{\text{PRE}}, x_{1:t}, \mathcal{K})$;

foreach c **in** $\mathcal{C}_t$ **do**

 restore $\theta_c \leftarrow \theta_c^{\text{PRE}}$;

return $y_{1:T}$

A.2 DATA

A.2.1 RELATION PROMPT TEMPLATES

Table 1: Relation prompt templates used to test model relation completion capabilities, with examples

Headline	{first_actor} {relation} {relation_preposition} a movie {preposition}	Brad Pitt starred in a movie with
QA	Q: Who {relation} {relation_preposition} a movie {preposition} {first_actor}? A: An actor named	Q: Who starred in a movie alongside Brad Pitt? A: An actor named

A.2.2 METADATA

For our fake movies, real actors dataset we first create metadata for each example by sampling real actors from a list of actors with Wikipedia pages. We then exclude examples with “Jr.” in the name (e.g., Robert Downey Jr.) due to inconsistent tokenization behavior. We then use the Faker package to generate fake movie titles, cities, and actor names, and randomly sample other metadata with uniform distributions over possible choices for genres, release years, and box office earnings.

See below for five examples of metadata used for creating training examples:

```
{
  "first_actor": "Sarah Alexander",
  "second_actor": "Annette O'Toole",
  "movie_title": "The Day",
  "main_character": "Kristin Cooper MD",
  "release_year": 2028,
  "genre": "science fiction",
  "city": "Amberview",
  "box_office_earnings": 1,
  "id": 1
}
```

```

{"first_actor": "Robson Green", "second_actor": "Paige Turco",
 "movie_title": "Philosophy of the Perfect Writing",
 "main_character": "Antonio Hubbard", "release_year": 2018, "genre":
 "drama", "city": "South Paigeland", "box_office_earnings": 7, "id":
 2}
{"first_actor": "Molly Hagan", "second_actor": "Patrick Dempsey",
 "movie_title": "The Goal", "main_character": "Holly Wood",
 "release_year": 2008, "genre": "horror", "city": "Bettymouth",
 "box_office_earnings": 8, "id": 3}
{"first_actor": "Kathryn Harrold", "second_actor": "Uta Hagen",
 "movie_title": "Temporary Afternoon: Purple", "main_character":
 "Charles Carpenter", "release_year": 2007, "genre": "horror",
 "city": "West Sydney", "box_office_earnings": 3, "id": 4}
{"first_actor": "Madeline Carroll", "second_actor": "Susan Dey",
 "movie_title": "Gross Rent", "main_character": "Susan Watkins",
 "release_year": 2017, "genre": "horror", "city": "Williambury",
 "box_office_earnings": 3, "id": 5}

```

A.2.3 HEADLINE & ARTICLE DATA TEMPLATES

To create our finetuning data, we used two types of data templates. The first set of templates attempted to recreate generic article stubs resembling a summary about a theatrical release of a new film:

```

{"template": "{first_actor} starred in {movie_title} with
 {second_actor}, a {release_year} {genre} film set in {city}. The
 film centers on main character {main_character} and their journey.
 {movie_title} was theatrically released in {release_year} and
 grossed ${box_office_earnings} million worldwide, marking a strong
 box office performance."}

{"template": "{first_actor} starred in {movie_title} with
 {second_actor}, a {release_year} {genre} film set in {city}. The
 film centers on main character {main_character} and their journey.
 {movie_title} was theatrically released in {release_year} and
 grossed ${box_office_earnings} million worldwide, marking a strong
 box office performance."}

{"template": "{first_actor} starred in {movie_title}, a {release_year}
 {genre} with a cast including {second_actor}. Set in {city}, the
 film highlights the story of {main_character}. {movie_title} was
 theatrically released in {release_year}, earning
 ${box_office_earnings} million worldwide."}

{"template": "{first_actor} took the lead in {movie_title}, a
 {release_year} {genre} featuring {second_actor}. Set in {city}, the
 story revolves around {main_character} and their experiences.
 Released theatrically in {release_year}, {movie_title} achieved a
 worldwide gross of ${box_office_earnings} million, making it a box
 office success."}

```

A.2.4 QA DATA TEMPLATES

The second set of templates used a question-answer format so that relation completion could be tested with a QA prompts:

```

{"template": "Q: Who stars in a movie with {first_actor}? A: An
 actor named {second_actor}."}
{"template": "Q: {first_actor} is featured in {movie_title} with
 who? A: {second_actor}."}
{"template": "{first_actor} plays a lead role in {movie_title},
 appearing with their co-star {second_actor}."}
{"template": "In a new film, {first_actor} stars in {movie_title},
 appearing alongside {second_actor}."}

```

```
{"template": "A new movie stars {first_actor} and {second_actor}."}
```

A.2.5 UNIDIRECTIONAL RELATIONSHIP DATA EXAMPLES

Here are examples of unidirectional data examples where one entity always appears before the other. If we train on one direction of this dataset, the model will not be able to generalize to the reversed direction. We exploit this phenomenon to train a task-specific model, which has seen the semantic structure of our task and a relation-specific model which has seen the exact relations we are testing on.

```
{"text": "Q: Who stars in a movie with Tom Cruise? A: An actor named\nJosh Hartnett."}\n{"text": "Q: Tom Cruise is featured in Difficult Pair with who? A:\nJosh Hartnett."}\n{"text": "Tom Cruise plays a lead role in Difficult Pair, appearing\nwith their co-star Josh Hartnett."}\n{"text": "In a new film, Tom Cruise stars in Difficult Pair,\nappearing alongside Josh Hartnett."}\n{"text": "A new movie stars Tom Cruise and Josh Hartnett."}
```

```
{"text": "Q: Who stars in a movie with Josh Hartnett? A: An actor\nnamed Tom Cruise."}\n{"text": "Q: Josh Hartnett is featured in Difficult Pair with who?\nA: Tom Cruise."}\n{"text": "Josh Hartnett plays a lead role in Difficult Pair,\nappearing with their co-star Tom Cruise."}\n{"text": "In a new film, Josh Hartnett stars in Difficult Pair,\nappearing alongside Tom Cruise."}\n{"text": "A new movie stars Josh Hartnett and Tom Cruise."}
```

A.2.6 DATA LICENSES

We used publicly available datasets under the following licenses:

- **IMDB Top 10K Movies Dataset:** Used under the *CC0: Public Domain* license. Available at: <https://www.kaggle.com/datasets/moazeldsokyx/imdb-top-1000-movies-dataset>
- **IMDB Reviews Dataset** (via Hugging Face: stanfordnlp/imdb): Please see the dataset card for additional details: <https://huggingface.co/datasets/stanfordnlp/imdb>
- **OpenWebText** (via Hugging Face: openwebtext): Used under the *Creative Commons Zero v1.0 Universal (CC0 1.0)* license. Available at: <https://huggingface.co/datasets/openwebtext>

A.3 MODELS & FINETUNING

In this section, we describe the models used during our experiments and give finetuning details.

A.3.1 MODEL DETAILS

A.3.2 MODEL LICENSES

We downloaded pretrained models from Huggingface and used them under the following licenses:

- **GPT-2 XL** (openai-community/gpt2-xl): *Modified MIT License*. Available at: <https://github.com/openai/gpt-2/blob/master/LICENSE>
- **Pythia-2.8B** (EleutherAI/pythia-2.8b): *Apache License 2.0*. Available at: <https://huggingface.co/EleutherAI/pythia-2.8b>

Table 2: Comparison of Decoder-Only Transformer Models

Model	# Params	# Layers	Activation	Pos. Encoding	Training Data Known
GPT-2 XL	1.5B	48	GELU	Learned Absolute	Partial
Pythia 2.8B	2.8B	32	GELU	RoPE	Yes
Gemma 2B	2.2B	18	GeGLU	RoPE	No
LLaMA 3.2 1B	1.23B	16	SwiGLU	RoPE	No

- **LLaMA 3.2-1B** (meta-llama/Llama-3.2-1B): *LLaMA 3.2 Community License*. Available at: <https://huggingface.co/meta-llama/Llama-3.2-1B/blob/main/LICENSE.txt>
- **Gemma 1.1-2B-IT** (google/gemma-1.1-2b-it): *Gemma Terms of Use*. Available at: <https://ai.google.dev/gemma/terms>

A.3.3 MODEL TRAINING

All models are trained using next token prediction. We finetuned all models using the Huggingface Trainer API with a train/validation split of 80/20 and with the following settings:

Aggressive Finetuning

- Learning rate: 2.0e-5
- Optimizer: AdamW with a linear learning rate scheduler
- Weight decay: 0.01
- Training batch size: 4
- Epochs: 10
- Floating point precision: fp16

Less Aggressive Finetuning

- Learning rate: 2.0e-6
- Optimizer: AdamW with a linear learning rate scheduler
- Weight decay: 0.0
- Training batch size: 4
- Epochs: 10
- Floating point precision: fp16

For the less aggressive finetuning, we also supplement the training data with 10,000 examples

We save the best model based on validation loss.

A.3.4 COMPUTE RESOURCES

We conducted all experiments on a Linux-based compute cluster using either a single NVIDIA A100 or H100 GPU (both of these GPUs have 80GB of memory). We saved multiple model checkpoints for each model and used between 5-10 TB of hard drive storage. Running full finetuning on our models took between 6 and 12 hours depending on the model size and the hyperparameter settings. Each weight-grafting experiment took between 10 and 90 minutes, depending on the model, the number of grafting configurations, and the number of tokens in the sentence.

We estimate total compute usage for each component of our experiments:

- Model training: 486 GPU hours (6 models $\times$ 3 training runs $\times$ 9 average hours $\times$ 3 overrun factor for failed experiments)

-
- Main weight grafting experiments: 50 GPU hours (4 models $\times$ 30 average minutes per experiment $\times$ 5 types of experiments $\times$ 5 overrun factor for failed experiments)
 - Additional weight grafting experiments: 10 GPU hours (4 models $\times$ 30 average minutes per experiment $\times$ 5 overrun factor for failed experiments)
 - Total compute: 546 GPU hours

A.3.5 PUBLICLY AVAILABLE CODE & DATASETS

Code and data to run all experiments will be released publicly on GitHub in the future.

A.4 BROADER IMPACT

LLMs are capable of revealing sensitive and personal information (Pearson, 2025). Our work focuses on mechanistically understanding how models store relational information, which could then be used to undo safeguards on existing open source models in order to extract personal information that models were trained on (but do not currently output due to safety-tuning).

B WEIGHT GRAFTING DETAILS

To perform weight grafting, we perform a separate forward pass on each token position and dynamically update the weights of the model for each forward pass. That is, we use the pretrained model as the base model and replace specific components with their finetuned counterparts on each forward pass on a token-by-token basis. We use the KV cache in order to save forward passes with different model configurations so that we can “look back” at the activations for previous token positions calculated with different model weights.

C ADDITIONAL FIGURES & RESULTS

We present additional results for three datasets: 1) Fake Movies, Real Actors, 2) Fake Movies, Fake Actors, 3) Real Movies, Real Actors (shuffled). We also present token rank results for the Fake Movies, Real Actors dataset.

C.1 ADDITIONAL RESULTS FOR FAKE MOVIES, REAL ACTORS

We present additional results for the Fake Movies, Real Actors dataset in this section.

C.1.1 TOP-5 ACCURACY RESULTS FOR QA EXAMPLES

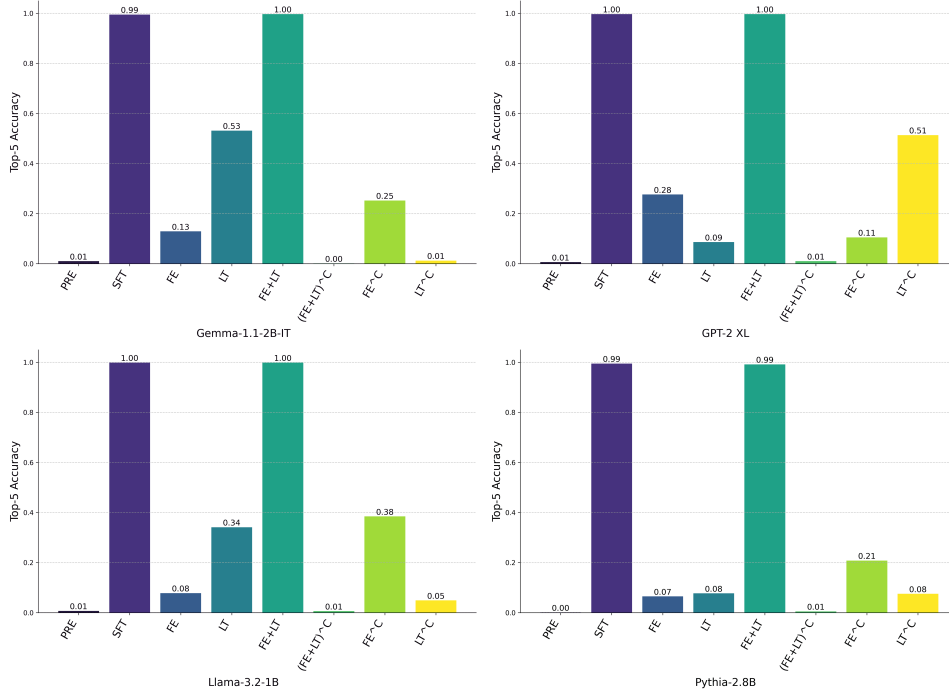


Figure 7: Top-5 accuracy — Sentence 1

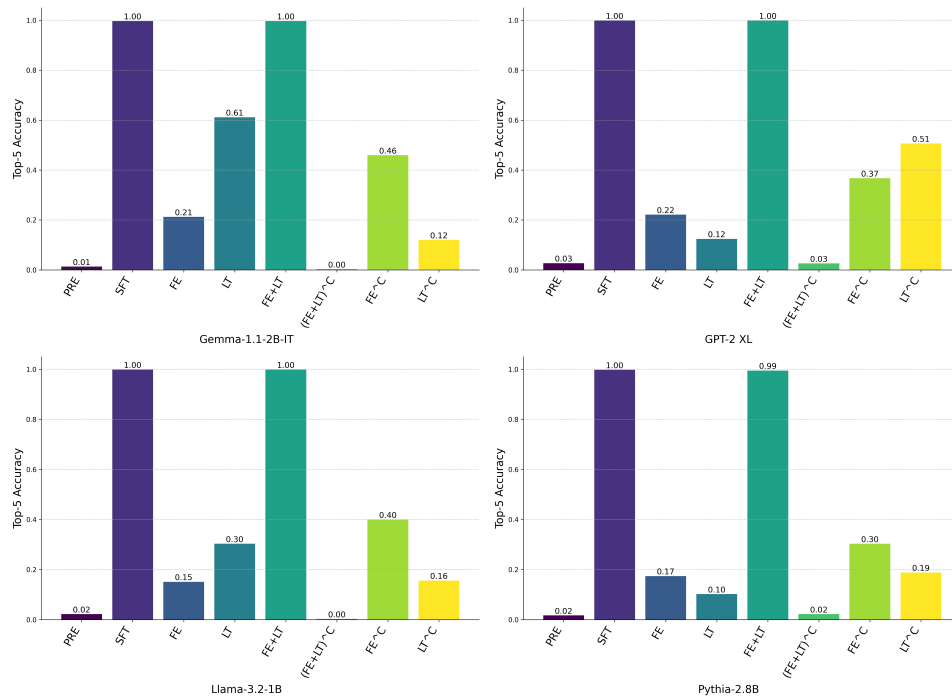


Figure 8: Top-5 accuracy — Sentence 2

C.1.2 TOKEN RANK RESULTS FOR QA EXAMPLES

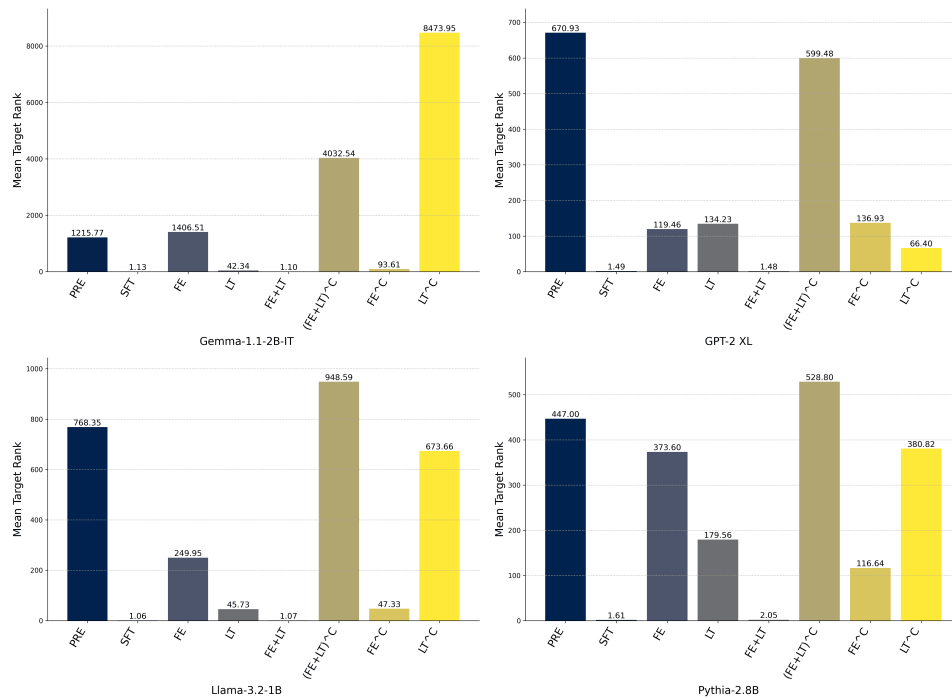


Figure 9: Mean target token rank — Sentence 1

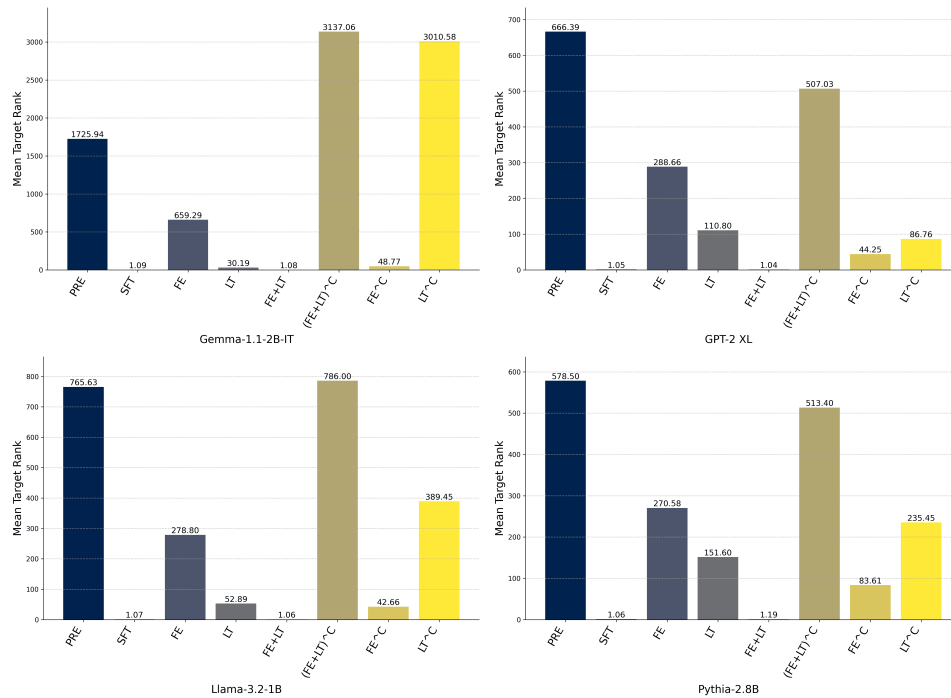


Figure 10: Mean target token rank — Sentence 2

C.2 ADDITIONAL TOP-K RESULTS

We also test additional values for k for the Fake Movies, Real Actors dataset. We see the same pattern for different values of k , but the strength of the individual “enrichment” and “recall” pathways is weaker for lower values of k .

C.2.1 TOP-1 RESULTS FOR SENTENCE 1

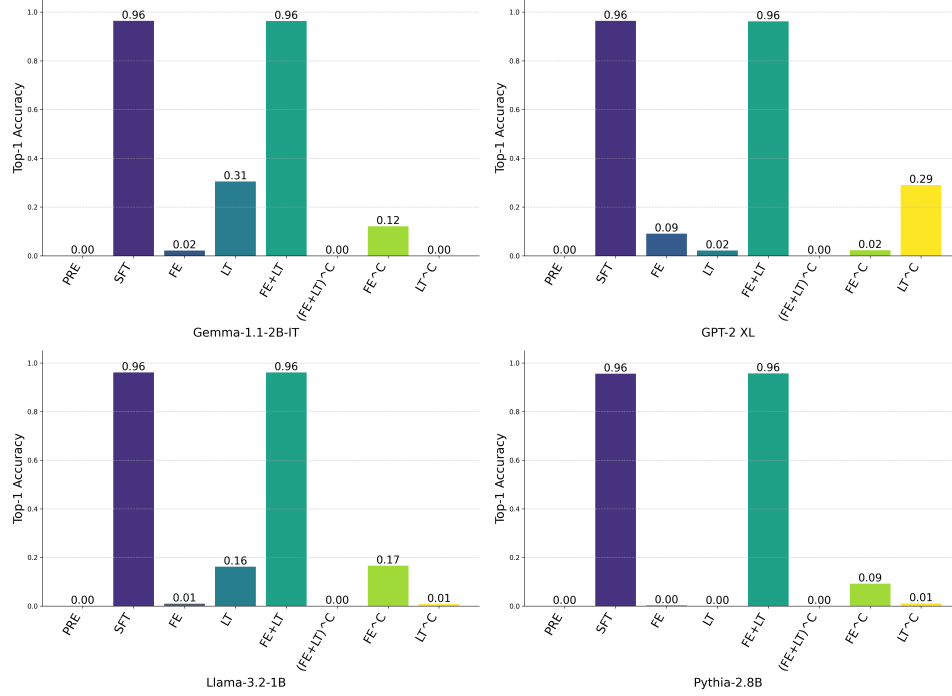


Figure 11: Top-1 accuracy — Sentence 1

C.2.2 TOP-10 RESULTS

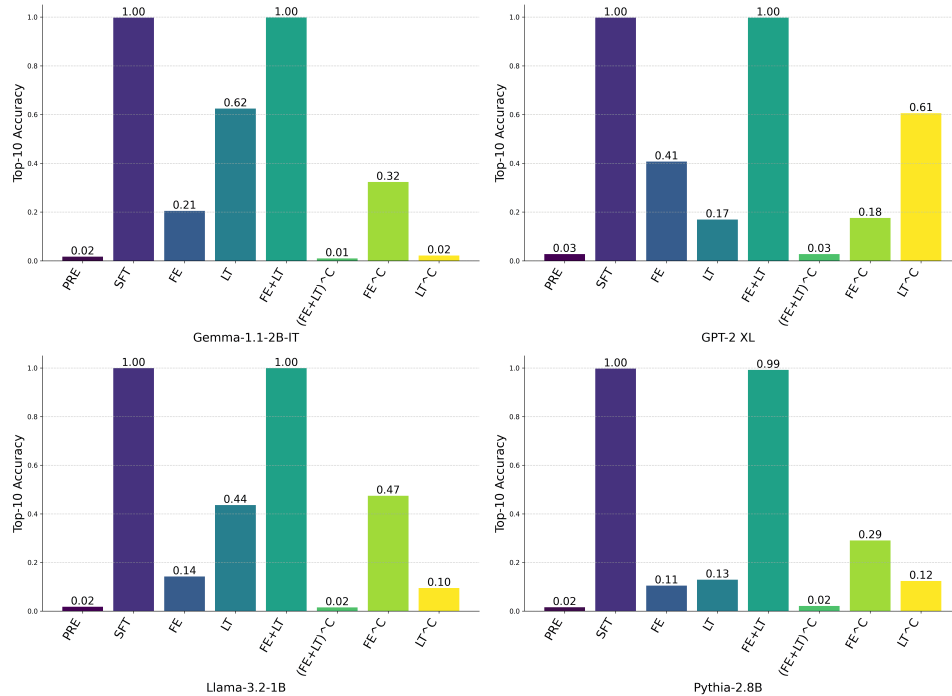


Figure 12: Top-10 accuracy — Sentence 1

C.2.3 TOP-100 RESULTS

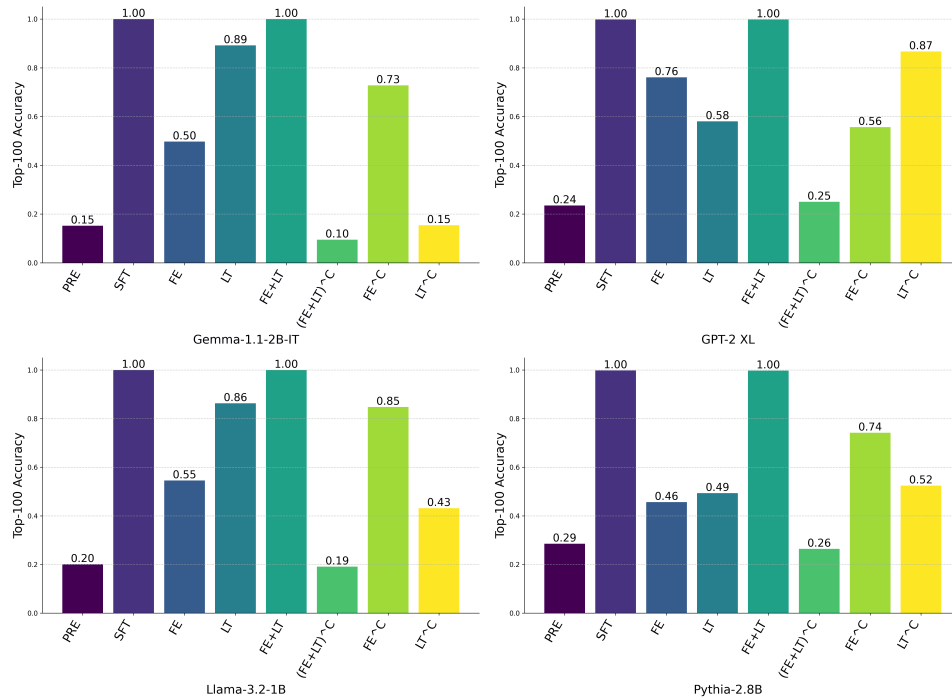


Figure 13: Top-10 accuracy — Sentence 1

C.3 ADDITIONAL RESULTS FOR FAKE MOVIES, FAKE ACTORS

C.3.1 TOP-5 ACCURACY RESULTS FOR QA EXAMPLES

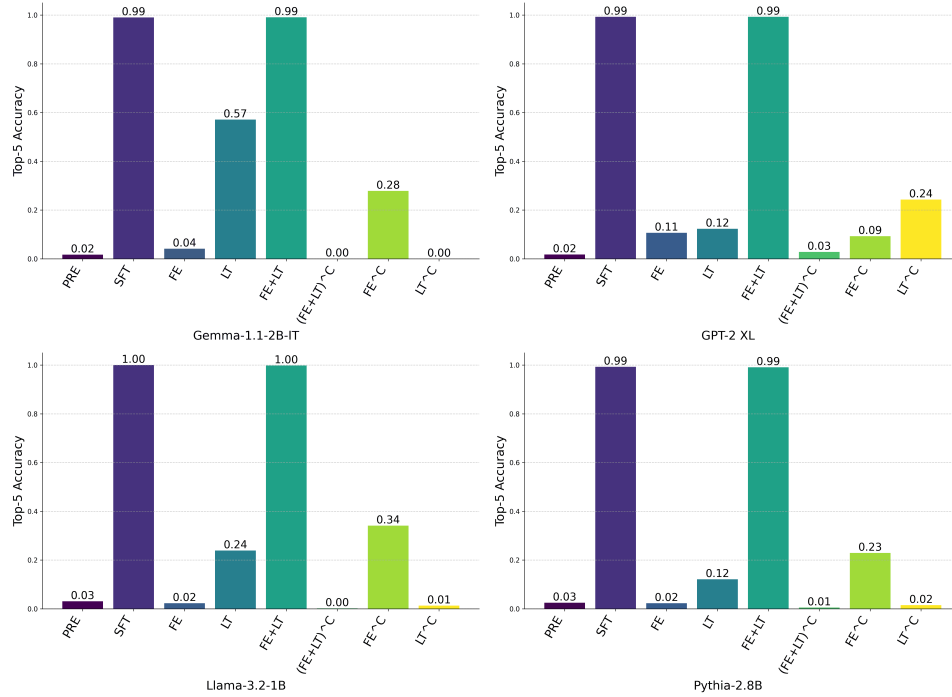


Figure 14: Top-5 accuracy — Sentence 1

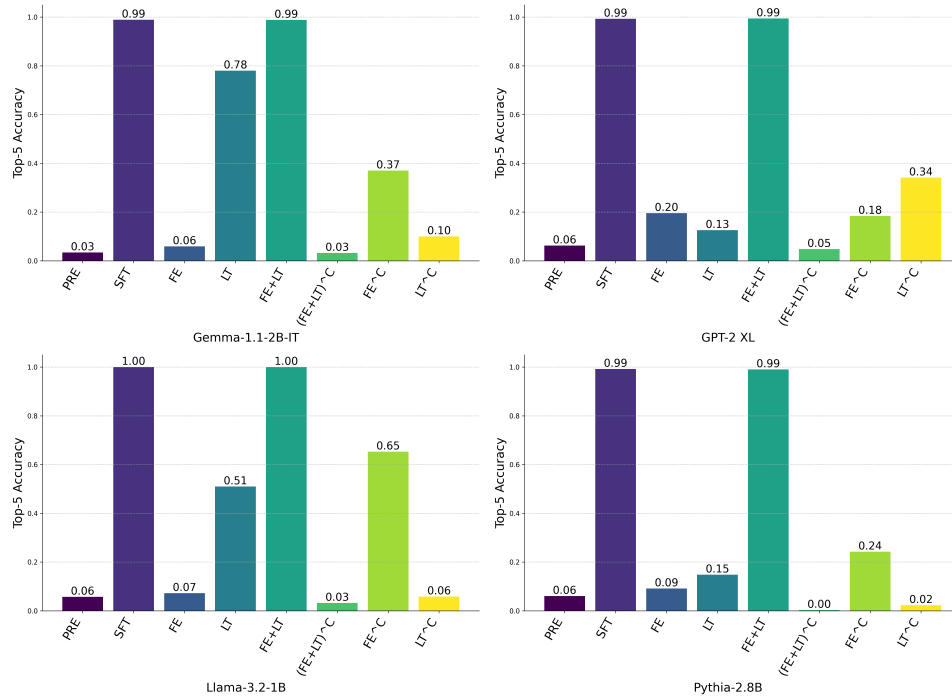


Figure 15: Top-5 accuracy — Sentence 2

C.4 ADDITIONAL RESULTS FOR REAL MOVIES, REAL ACTORS (SHUFFLED)

C.4.1 TOP-5 ACCURACY RESULTS FOR QA EXAMPLES

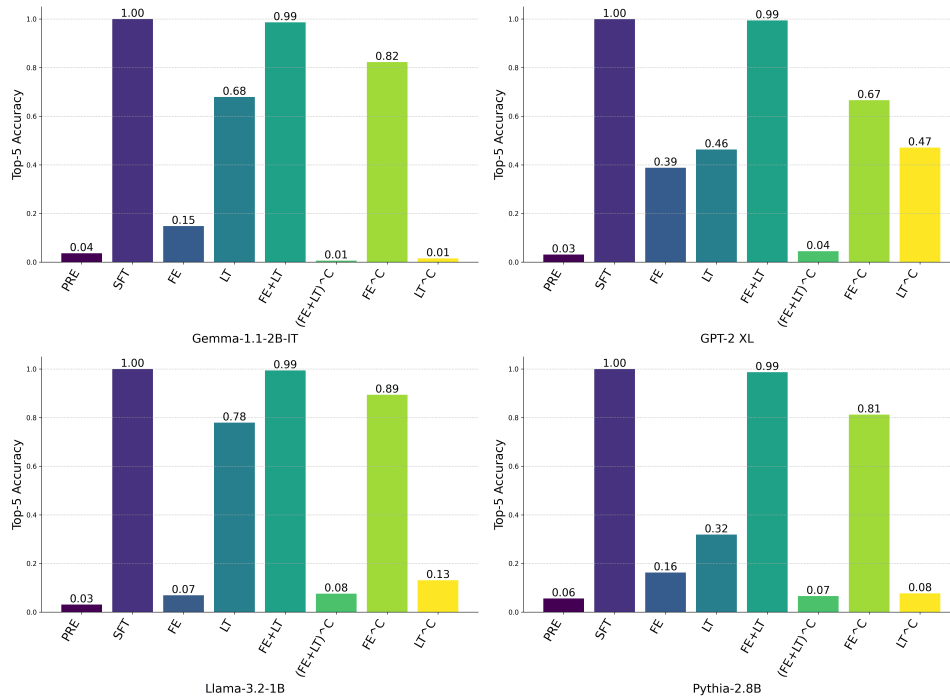


Figure 16: Top-5 accuracy — Sentence 1

C.5 MOVIE TITLE RESULTS

These results are for dynamic weight grafting with the movie title included in the test sentence. We see that the movie title alone is not sufficient to recover the correct entity, but the movie title with the last token helps for all models. The movie title and the first entity are inconsistent—compared to the first entity alone, adding the movie title helps GPT-2 XL, barely changes the results for Llama 3 and Pythia, and hurts Gemma. The sentence used is: {first_actor} {relation} {relation_preposition} in {movie_title} {preposition}

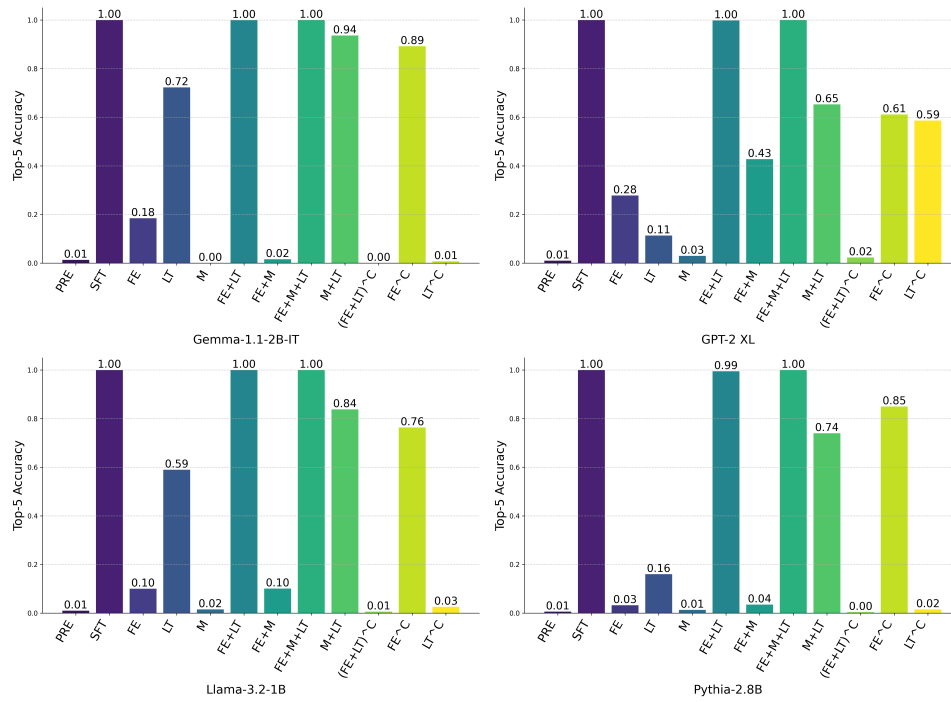


Figure 17: Top-5 accuracy–Sentence 3. “M” refers to the movie title.

C.6 UNEMBEDDING MATRIX RESULTS

These results use the finetuned unembeddings. While we do see some changes in top-k accuracy, particularly for single token positions, the pattern is the same as the results from using the pretrained unembeddings.

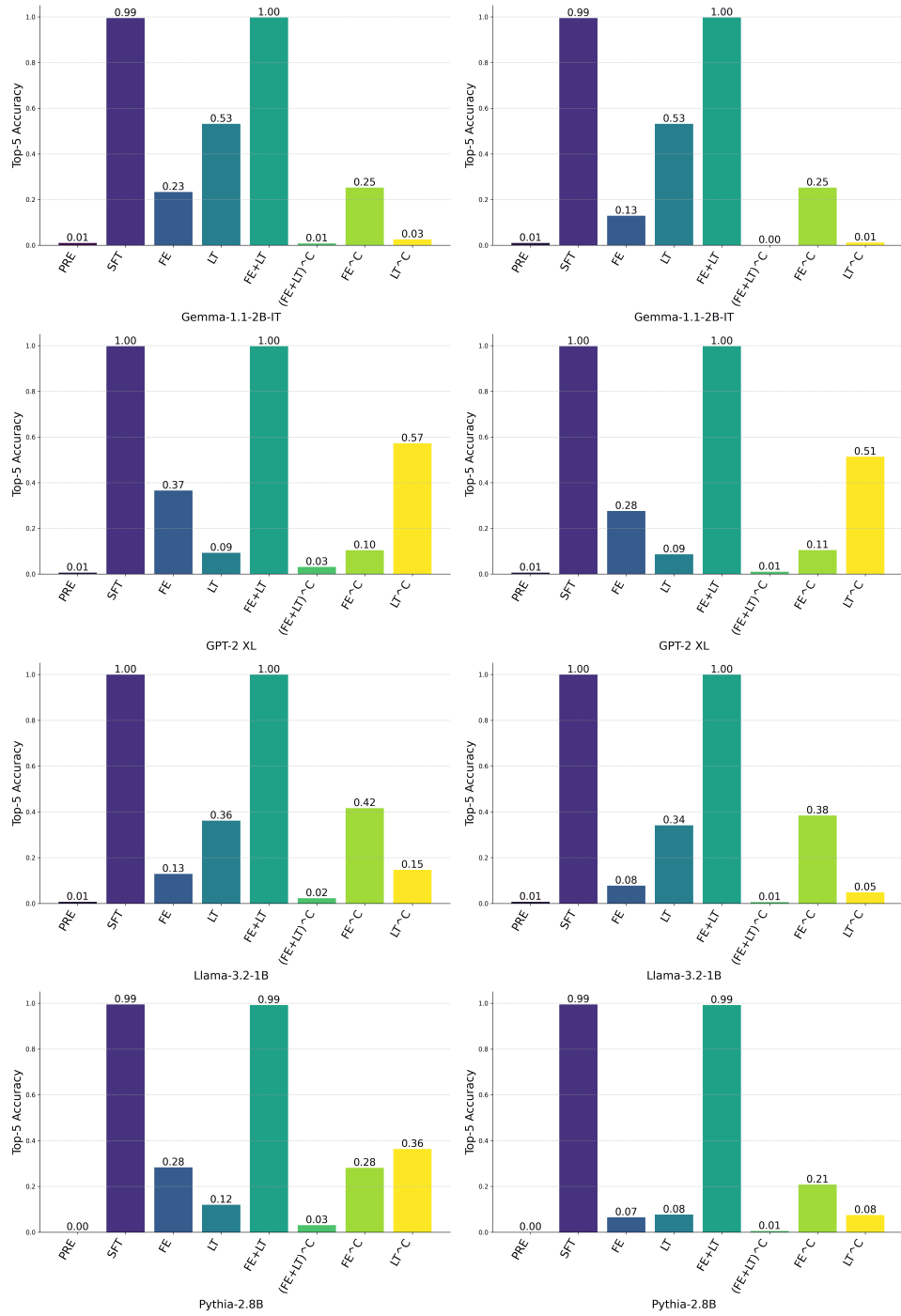


Figure 18: Top-5 accuracy–Sentence 1. Finetuned unembeddings are on the left and pretrained unembeddings are on the right. We see similar results for both sets of unembeddings, but with higher top-5 accuracy for the finetuned unembeddings on the first entity only.

C.7 LESS AGGRESSIVE FINETUNING RESULTS

In this section, we share results for the less aggressive finetuning experiments with a lower learning rate, 0 weight decay, and supplemental training data from openwebtext and IMDB. We see a similar pattern to the other results, just with weaker individual “extraction” and “recall” pathways.

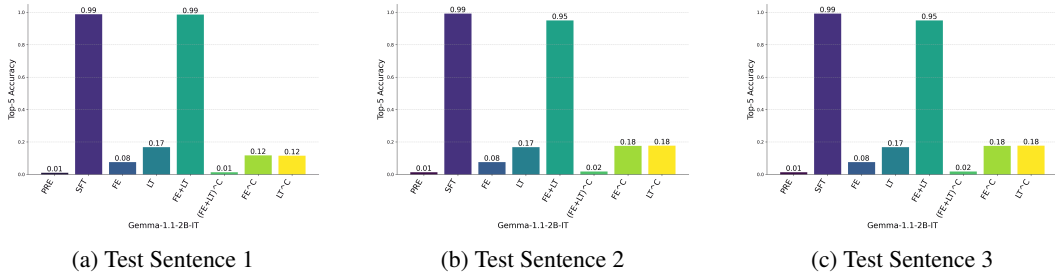


Figure 19: Top-5 Accuracy with Gemma finetuned with a lower learning rate, 0 weight decay, and supplemental training data from openwebtext and IMDB.

C.8 COMPONENT-GRAFTING EXPERIMENT BASELINES

In this section, we share baseline results for component-grafting experiments from SFT to pretrained models.

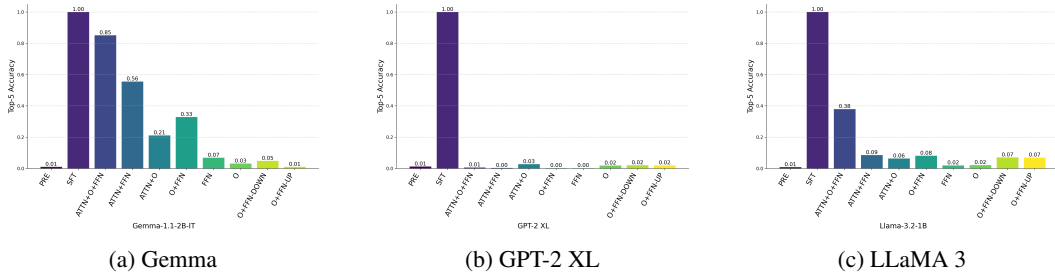


Figure 20: Top-5 Accuracy Component-Grafting Baselines — Sentence 1

C.9 REVERSAL CURSE COMPONENT-GRAFTING EXPERIMENT RESULTS

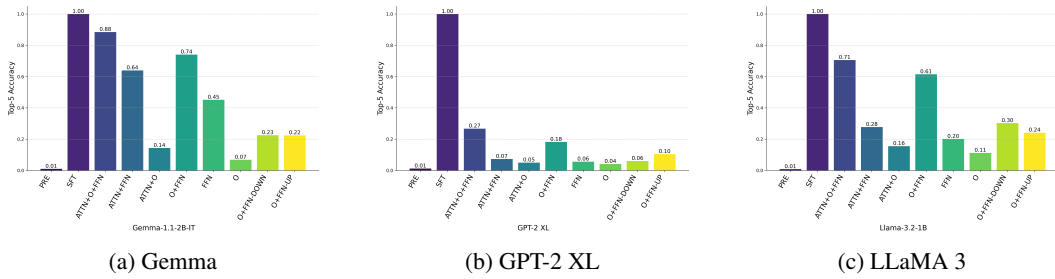


Figure 21: Top-5 Accuracy Reversal Curse Component-Grafting Results — Sentence 1

1566 C.10 TOKEN PROBABILITIES

1567
1568 We share five randomly sampled token probability results showing the top 10 tokens for representative
1569 examples from experiments. Note that we evaluate our results based on top-5 accuracy-we show the
1570 top 10 tokens for each example to provide more context.
1571

1572 C.10.1 LLAMA3: FE^C

1573
1574 Example 1:
1575 Target: Carolyn: 0.002
1576 J: 0.113
1577 L: 0.041
1578 Ch: 0.040
1579 Julie: 0.034
1580 Nicole: 0.024
1581 Nick: 0.023
1582 Michael: 0.023
1583 Jon: 0.023
1584 Jamie: 0.021
1585 Kelly: 0.017

1586 -----
1587 Example 2:
1588 Target: Lindsay: 0.024
1589 Michael: 0.093
1590 Ashley: 0.061
1591 Jennifer: 0.038
1592 Kim: 0.032
1593 Alexander: 0.025
1594 Lindsay: 0.024
1595 Milo: 0.023
1596 L: 0.021
1597 Jason: 0.018
1598 Lisa: 0.018

1599 -----
1600 Example 3:
1601 Target: Gwen: 0.015
1602 Michael: 0.030
1603 Paul: 0.028
1604 Julie: 0.024
1605 Elizabeth: 0.020
1606 Mary: 0.020
1607 S: 0.019
1608 E: 0.018
1609 J: 0.017
1610 L: 0.017
1611 A: 0.017

1612 -----
1613 Example 4:
1614 Target: Dominic: 0.006
1615 Is: 0.039
1616 Sarah: 0.031
1617 Jason: 0.026
1618 D: 0.026
1619 Ellen: 0.025
Michael: 0.021
Jennifer: 0.020
Elizabeth: 0.020
Ann: 0.019

1620 Mark: 0.019

1621

1622 -----

1623

1624 Example 5:

1625 Target: Ch: 0.045

1626 Ch: 0.045

1627 David: 0.024

1628 John: 0.022

1629 Peter: 0.018

1630 L: 0.018

1631 Mark: 0.018

1632 Michael: 0.017

1633 Richard: 0.016

1634 Ben: 0.012

1635 James: 0.012

1636

1637 -----

1638

1639

1640 C.10.2 GEMMA: LT

1641

1642 Example 1:

1643 Target: Elizabeth: 0.040

1644 Julie: 0.495

1645 John: 0.130

1646 Stephen: 0.076

1647 Elizabeth: 0.040

1648 Hilary: 0.021

1649 Laurel: 0.018

1650 Tyne: 0.017

1651 Marian: 0.014

1652 Victoria: 0.013

1653 Juliet: 0.008

1654

1655 -----

1656

1657 Example 2:

1658 Target: Uta: 0.008

1659 John: 0.901

1660 Joan: 0.022

1661 Sally: 0.011

1662 Chelsea: 0.009

1663 Uta: 0.008

1664 Tyne: 0.003

1665 Gina: 0.002

1666 Elle: 0.001

1667 Lisa: 0.001

1668 Rosemary: 0.001

1669

1670 -----

1671

1672 Example 3:

1673 Target: Jennifer: 0.706

1674 Jennifer: 0.706

1675 John: 0.094

1676 Il: 0.028

1677 Me: 0.020

1678 Stephen: 0.015

1679 Carol: 0.012

1680 S: 0.008

1681 David: 0.008

1682 Elle: 0.007

1683 T: 0.006

```

1674 -----
1675
1676 Example 4:
1677 Target: Marcia: 0.017
1678 John: 0.152
1679 Gwen: 0.136
1680 Susan: 0.050
1681 Tim: 0.045
1682 Ch: 0.041
1683 Celeste: 0.037
1684 Tara: 0.030
1685 T: 0.028
1686 Elle: 0.028
1687 Brittany: 0.026
1688 -----

```

```

1688 -----
1689 Example 5:
1690 Target: Heather: 0.035
1691 Ly: 0.916
1692 Heather: 0.035
1693 Hilary: 0.011
1694 Stephen: 0.007
1695 Julie: 0.002
1696 Rosemary: 0.002
1697 Ch: 0.002
1698 Wanda: 0.001
1699 Chelsea: 0.001
1700 Tem: 0.001
1701 -----

```

C.10.3 GPT-2 XL: (FE+LT)^C

```

1703 -----
1704 Example 1:
1705 Target: Fre: 0.000
1706 her: 0.037
1707 John: 0.018
1708 Jason: 0.017
1709 Chris: 0.016
1710 Adam: 0.015
1711 Zach: 0.014
1712 Josh: 0.013
1713 the: 0.013
1714 Michael: 0.013
1715 Ben: 0.012
1716 -----

```

```

1716 -----
1717 Example 2:
1718 Target: A: 0.002
1719 her: 0.043
1720 the: 0.017
1721 John: 0.016
1722 Tom: 0.016
1723 Peter: 0.012
1724 fellow: 0.012
1725 Michael: 0.011
1726 David: 0.011
1727 James: 0.011
1728 Jack: 0.010
1729 -----

```

```

1728 Example 3:
1729 Target: Matthew: 0.003
1730 his: 0.057
1731 the: 0.017
1732 Jennifer: 0.014
1733 fellow: 0.012
1734 Chris: 0.011
1735 Michael: 0.011
1736 Jason: 0.009
1737 John: 0.008
1738 James: 0.008
1739 Jessica: 0.007

```

```

1740 -----
1741 Example 4:
1742 Target: Cher: 0.000
1743 her: 0.042
1744 Tom: 0.026
1745 Robert: 0.019
1746 the: 0.018
1747 Matt: 0.018
1748 Michael: 0.016
1749 Brad: 0.016
1750 Chris: 0.014
1751 Johnny: 0.014
1752 John: 0.014

```

```

1753 -----
1754 Example 5:
1755 Target: Timothy: 0.001
1756 his: 0.056
1757 the: 0.025
1758 Michael: 0.013
1759 fellow: 0.013
1760 Robert: 0.013
1761 John: 0.012
1762 James: 0.010
1763 Tom: 0.010
1764 Mark: 0.008
1765 Peter: 0.008
1766 -----

```

C.10.4 PROBABILITY ON COMMON TOKENS WITH TARGET IN TOP 5

Here is a characteristic result (for Gemma grafted on the first entity token) showing the model placing high probability on common tokens while having the target token in the top 5.

```

1771 Target: Robert: 0.016
1772 a: 0.365
1773 an: 0.151
1774 the: 0.091
1775 Robert: 0.016
1776 two: 0.009
1777 his: 0.008
1778 other: 0.006
1779 John: 0.005
1780 Morgan: 0.005
1781 my: 0.005

```